EARTH AND PLANETARY SCIENCE LETTERS 4 (1968) 448-450. NORTH-HOLLAND PUBLISHING COMP., AMSTERDAM

THE PALEOMAGNETISM AND GEOLOGY OF NORTH-WESTERN ICELAND

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Received 14 June 1968

Preliminary paleomagnetic observations from the Tertiary North-Western peninsula of Iceland are described. Their results are shown to yield information on local geological problems such as stratigraphy, absolute ages, eruption processes, paleoclimates and heat flow. The rocks in the area generally have stable directions of remanence, but the dispersion of these directions is remarkably large.

In contrast to most other parts of Iceland, the geology of the island's North-Western region is so far unmapped. This region, built almost entirely from flood basalts, may contain the oldest exposed rocks in Iceland, but no radiometric data are available from there [1]. Interbasaltic lignites and leaf beds occur, indicating a warm-temperature climate, but paleontologists have not been able to place these more accurately in time then "Eocene to Miocene" [2]. Some authors have claimed that the lignites form a single horizon in the region, while others assume that they present small oases which appeared sporadically in space and time.

Evidence on the nature of the lignite beds, and on their age, has now been obtained from the results of a paleomagnetic survey of the North-Western peninsula [3]. Particular attention was paid to the area between the Skutuls and Onundar fjords (near 66°N, 23 1/2°W), where the Botn lignite mine [4] occurs. The survey confirms the view [5] that the Botn lignite is an outcrop of an extensive horizon of lignite, tuff and limonitic clay: at six localities in the area (20 × 20 km) this horizon was found to be associated with one to six "normally"magnetized lavas within a "reverse" zone. Throughout the peninsula, such zones of alternating magnetic polarity occur, averaging about 20 lavas (200 m) in thickness. As the geological succession through Western and North-Western Iceland appears to be uniform and not much different from that of Eastern Iceland, it is concluded that these zones do not represent the long Lower Tertiary magnetic epochs dated elsewhere [6, p. 28]. The North-Western peninsula thus cannot be older than Upper Oligocene (30 m.y.).

The magnetic properties of 175 orientated samples from 60 lavas in the Skutuls-Önundar area were investigated in detail. (The notation here is that of Irving [7]). Their low-field susceptibilities ranged from 0.4 to 3 $G.Oe^{-1} \cdot 10^{-3}$, and their intensities of remanence (after a.c. demagnetization at 85 Oe peak field) from 0.4 to 4 $G.10^{-3}$. These properties are presumably due to titanomagnetite grains, which in the samples studied had an average diameter of 15μ. At magnifications $>$ 400x, exsolution structure was frequently seen in the grains. No obvious correlations were found between any magnetic or other properties, except that fine-grained, compact, fresh-looking lavas tended to have high susceptibilities and large components of viscous remanence (10–50% of their NRM). This fact may be relevant to the interpretation of local geomagnetic surveys. The viscous remanence could usually be eliminated by 50–120 Oe peak a.c. field, leaving a stable direction of primary TRM.

From the TRM, usually that measured after 150 Oe treatment, an average paleofield direction was found for each of the 60 lavas. The mean uncertainty

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(θ^{o}_{63}) in each direction, including possible errors in the measurement of regional dip, is only about 8°. The lavas span four geomagnetic epochs. Both polarities occurred in one lava, but this fact was not found to be due to any reproducible self-reversal. Unit weight being given to each lava direction, regardless of polarity, a grand average paleofield direction of $D = 0.5^{o}$, $I = +73.0^{o}$ was obtained, with $N = 60, R = 55.42, k = 12.87, \delta = 22.5^{o}, \alpha_{95} = 5.3^{o}$. On the "central dipole" assumption, this field corresponds to a geomagnetic pole at 82.5°N, 155°E, with $\delta p_{95} = 8.4^{o}$, $\delta m_{95} = 9.3^{o}$. This pole position is similar to those usually quoted for Oligo-Miocene rocks.

The large value of δ obtained here may be typical of the Neogene geomagnetic field in Iceland, c.f. the results of Smith [8] (which may, however, not have been corrected for dips of strata of the order of 5°), and related to the high frequency of reversals at the time. It follows that in Iceland and in locations of similar age and latitude, any paleomagnetic survey should include some 200 independent directions from at least two magnetic epochs, if their mean pole is to be reliably located within a 5° radius of 95% confidence.

An outstanding problem in the geology of Iceland has been that of the origin of the thin red or brown interbasaltic beds, which occur in most parts of the country. This problem is reflected in their being called "dust beds" by some authors and "laterites" by others. Microscope examination of several samples from such beds [3] shows that they contain small grains of magnetite, apparently in the process of being dissolved into an amorphous red material. The magnetite and some hematite or goethite in the red material have identical directions of remanence. Both these observations are in agreement with the view of Einarsson [9] that the red beds consist chiefly of clay minerals, formed from volcanic dust by chemical weathering in a humid climate and subsequently baked by the overlying lava.

Still another geological problem in Iceland concerns the causes of the high present-day heat flow in the Tertiary areas of the country. In a thorough analysis of the available data, Bödvarsson [10] assumed, as a boundary condition, that prior to Pliocene erosion the temperature at a certain stratigraphic level above the Both lignites reached 500°C. In order to improve on this estimate, stepwise thermal demagnetization was carried out (in air) on samples from fifteen lavas from North-Western Iceland [3]. Magnetic properties and directions of remanence varied widely between these samples; prior to heating, viscous magnetization was removed by a low a.c. field. In most of the samples, the direction of remanence did not change by more than a few degrees on heating to the Curie point (280–560°C), while a significant proportion of the intensity of remanence would be lost already on heating to 150°C. Further tests were applied to those samples which showed no significant chemical changes during heating, establishing the approximate validity of Thellier's law of PTRM in their case. It was concluded that the lava pile in Iceland has in general not been heated to more than 120°C since its formation. This conclusion can be tested further by a study of baked margins in the lavas. If confirmed, thermomagnetic methods could be applied to other aspects of the paleo-heat flow in Iceland, such as that of maximum temperatures reached around volcanic centres.

Results from the paleomagnetic survey described above do not confirm earlier suggestions that in Iceland either the lignite beds, or eruptions of porphyritic lavas, are connected with unusually long periods of volcanic quiescence. Furthermore, field evidence obtained during the survey does not support the theory that the interbasaltic beds originated as poorly consolidated eolian deposits in a dry lava desert. Other, more tentative, conclusions can be drawn from the survey, but recording of these, and further paleomagnetic work in Western Iceland, should succeed stratigraphic mapping of the area.

ACKNOWLEDGEMENTS

I thank the Science Fund of Iceland for financial support and Professor S.K.Runcorn for use of facilities at the Department of Geophysics, University of Newcastle upon Tyne, England. I also thank Dr. H.C. Noltimier for encouragement and advice in laboratory work.

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